

INDUSTRY PROBLEM-SOLVING TASK

Java AWT, Applets, Event Handling and Layout Managers

Name of the Student	V.Bala Siva Prasad Reddy	Register No.	192424316
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Faculty	Dr. Sivasankar C	Duration	Lab Session

INDUSTRY TASK 1: Employee Performance & Payroll Dashboard

Problem Statement: A software company wants to develop a simple Employee Performance and Payroll Management System using Java AWT. The application should allow the HR team to enter employee details, process salary information, and generate a structured payroll report.

Functional Requirements

- Develop an AWT-based graphical user interface.
- Accept Employee ID, Employee Name, Department, Designation, Basic Salary, Working Days, and Performance Rating.
- Use suitable AWT components such as Label, TextField, Choice, Checkbox/CheckboxGroup, Button, TextArea, and List.
- Calculate Gross Salary, Performance Bonus, and Net Salary using suitable business rules defined by the student.
- Display the complete employee payroll report in a List or TextArea when Generate Report is clicked.
- Use GridBagLayout or BorderLayout for arranging the interface.
- Implement button events using ActionListener.
- Provide Submit/Generate Report and Reset functionality.
- Validate important inputs such as empty fields and numeric salary values.

Industry Context: The application simulates an HR payroll module used to capture employee information, perform basic payroll processing, and present a report for HR verification.

Expected Deliverables

- Java source code.
- Working GUI application with event handling.
- Sample input and output.
- Screenshot of the final application.
- Brief explanation of components, layout manager, and event handling used.

CODE:

```
import java.awt.*;
import java.awt.event.*;

class EmployeePayroll extends Frame implements ActionListener
{
    Label l1,l2,l3,l4,l5,l6,l7;
    TextField t1,t2,t3,t4,t5;
    Choice c1;
    Checkbox r1,r2,r3,r4;
    CheckboxGroup cg;
    Button b1,b2;
    TextArea ta;
    List list;

    EmployeePayroll()
    {
        Frame f = new Frame("Employee Performance & Payroll Dashboard");

        l1 = new Label("Employee ID");
        l2 = new Label("Employee Name");
        l3 = new Label("Department");
        l4 = new Label("Designation");
        l5 = new Label("Basic Salary");
        l6 = new Label("Working Days");
        l7 = new Label("Performance Rating");

        t1 = new TextField();
        t2 = new TextField();
        t3 = new TextField();
        t4 = new TextField();
        t5 = new TextField();

        c1 = new Choice();
        c1.add("IT");
        c1.add("HR");
        c1.add("Finance");
        c1.add("Marketing");

        cg = new CheckboxGroup();

        r1 = new Checkbox("Excellent",cg,false);
        r2 = new Checkbox("Good",cg,false);
        r3 = new Checkbox("Average",cg,false);
        r4 = new Checkbox("Poor",cg,false);

        b1 = new Button("Generate Report");
        b2 = new Button("Reset");

        ta = new TextArea();
        list = new List();

        f.setLayout(new GridBagLayout());
        GridBagConstraints g = new GridBagConstraints();

        g.fill = GridBagConstraints.HORIZONTAL;
        g.insets = new Insets(5,5,5,5);

        g.gridx=0; g.gridy=0;
        f.add(l1,g);
```

```

g.gridx=1;
f.add(t1,g);

g.gridx=0; g.gridy=1;
f.add(l2,g);
g.gridx=1;
f.add(t2,g);

g.gridx=0; g.gridy=2;
f.add(l3,g);
g.gridx=1;
f.add(c1,g);

g.gridx=0; g.gridy=3;
f.add(l4,g);
g.gridx=1;
f.add(t3,g);

g.gridx=0; g.gridy=4;
f.add(l5,g);
g.gridx=1;
f.add(t4,g);

g.gridx=0; g.gridy=5;
f.add(l6,g);
g.gridx=1;
f.add(t5,g);

g.gridx=0; g.gridy=6;
f.add(l7,g);
g.gridx=1;

Panel p = new Panel();
p.add(r1);
p.add(r2);
p.add(r3);
p.add(r4);
f.add(p,g);

g.gridx=0; g.gridy=7;
f.add(b1,g);

g.gridx=1;
f.add(b2,g);

g.gridx=0; g.gridy=8;
g.gridwidth=2;
f.add(ta,g);

b1.addActionListener(this);
b2.addActionListener(this);

f.setSize(600,600);
f.setVisible(true);

f.addWindowListener(new WindowAdapter()
{
    public void windowClosing(WindowEvent e)
    {
        f.dispose();
    }
}

```

```

    });
}

public void actionPerformed(ActionEvent e)
{
    if(e.getSource()==b1)
    {
        if(t1.getText().equals("") ||
           t2.getText().equals("") ||
           t3.getText().equals("") ||
           t4.getText().equals("") ||
           t5.getText().equals(""))
        {
            ta.setText("Please fill all fields.");
            return;
        }

        try
        {
            double basic = Double.parseDouble(t4.getText());
            int days = Integer.parseInt(t5.getText());

            if(basic <= 0 || days <= 0 || days > 31)
            {
                ta.setText("Enter valid salary and working days.");
                return;
            }

            String rating = cg.getSelectedCheckbox().getLabel();

            double hra = basic * 0.20;
            double da = basic * 0.10;
            double gross = basic + hra + da;

            double bonus = 0;

            if(rating.equals("Excellent"))
                bonus = basic * 0.20;
            else if(rating.equals("Good"))
                bonus = basic * 0.10;
            else if(rating.equals("Average"))
                bonus = basic * 0.05;
            else
                bonus = 0;

            double net = gross + bonus;

            ta.setText("");

            ta.append("EMPLOYEE PAYROLL REPORT\n");
            ta.append("-----\n");
            ta.append("Employee ID      : " + t1.getText() + "\n");
            ta.append("Employee Name   : " + t2.getText() + "\n");
            ta.append("Department      : " + c1.getSelectedItem() + "\n");
            ta.append("Designation     : " + t3.getText() + "\n");
            ta.append("Basic Salary    : " + basic + "\n");
            ta.append("Working Days    : " + days + "\n");
            ta.append("Performance     : " + rating + "\n");
            ta.append("-----\n");
            ta.append("HRA              : " + hra + "\n");
            ta.append("DA               : " + da + "\n");

```

```

        ta.append("Gross Salary      : " + gross + "\n");
        ta.append("Performance Bonus : " + bonus + "\n");
        ta.append("Net Salary       : " + net + "\n");
        ta.append("-----\n");

        list.add(t1.getText() + " - " + t2.getText() + " - " + net);
    }
    catch(Exception ex)
    {
        ta.setText("Please enter valid numeric values.");
    }
}

if(e.getSource()==b2)
{
    t1.setText("");
    t2.setText("");
    t3.setText("");
    t4.setText("");
    t5.setText("");
    ta.setText("");
    cg.setSelectedCheckbox(null);
    c1.select(0);
}

}

public static void main(String args[])
{
    new EmployeePayroll();
}

}

```

OUTPUT:

Employee Performance & Payroll Dashboard

Employee ID: E101

Employee Name: Bala

Department: IT

Designation: Developer

Basic Salary: 50000

Working Days: 26

Performance Rating: ☒ Excellent ☐ Good ☐ Average ☐ Poor

Basic Salary : 50000.0
 Working Days : 26
 Performance : Excellent

 HRA : 10000.0
 DA : 5000.0
 Gross Salary : 65000.0
 Performance Bonus : 10000.0
 Net Salary : 75000.0

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INDUSTRY TASK 2: Online Food Ordering and Billing System

Problem Statement: A restaurant wants to develop a simple Food Ordering and Billing Application for its counter staff. The application should allow customers to select food items, specify quantities, and generate a final bill.

Functional Requirements

- Develop the application using Java AWT.
- Create fields for Customer Name and Table Number.
- Provide Food Category, Food Item, and Quantity selection using appropriate AWT components.
- Provide Add Item, Generate Bill, Clear, Next, and Previous buttons.
- Display selected food items in a List or TextArea.
- Calculate item-wise cost, Subtotal, GST, and Final Bill Amount.
- Implement button click events using ActionListener.
- Use CardLayout to create separate screens for Customer/Order Details, Food Selection, and Bill/Order Summary.
- Validate quantity and ensure that an item is selected before adding it to the order.

Industry Context: The application represents a simplified restaurant POS (Point of Sale) system used for taking customer orders, processing prices, and generating a customer bill.

Expected Deliverables

- Java source code.
- Working multi-card AWT interface.
- Sample order and generated bill.
- Screenshot of each card/screen.
- Brief explanation of CardLayout, AWT components, and event handling.

Evaluation Rubric

Criterion	Weightage	Marks / Remarks
Problem Understanding	15%	
GUI Components and Layout	20%	
Event Handling	20%	
Functional Correctness	25%	
Input Validation and User Experience	10%	
Program Explanation / Viva	10%	

Student Instructions

1. The solution must be implemented in Java using the concepts covered in the laboratory syllabus.
2. Students should demonstrate compilation, execution, and output during the practical session.
3. Use meaningful variable names and appropriate comments.
4. Students must be able to explain the selected AWT components, layout manager, event handling mechanism, and processing logic.

CODE:

```
import java.awt.*;
import java.awt.event.*;

class FoodOrdering extends Frame implements ActionListener
{
    CardLayout cl;
    Panel cards;

    TextField name, table, qty;
    Choice category, item;
    List orderList;
    TextArea bill;

    Button next1, next2, previous1, previous2;
    Button add, generate, clear;

    double subtotal = 0;

    FoodOrdering()
    {
        Frame f = new Frame("Online Food Ordering and Billing System");

        cl = new CardLayout();
        cards = new Panel(cl);

        Panel p1 = new Panel(new GridLayout(4, 2, 10, 10));

        Label l1 = new Label("Customer Name");
        Label l2 = new Label("Table Number");

        name = new TextField();
        table = new TextField();

        next1 = new Button("Next");

        p1.add(l1);
        p1.add(name);

        p1.add(l2);
        p1.add(table);

        p1.add(new Label(""));
        p1.add(next1);

        Panel p2 = new Panel(new GridLayout(7, 2, 10, 10));

        Label l3 = new Label("Food Category");
        Label l4 = new Label("Food Item");
        Label l5 = new Label("Quantity");

        category = new Choice();
        category.add("Main Course");
        category.add("Fast Food");
        category.add("Beverages");

        item = new Choice();
        item.add("Biriyani - 220");
```

```

item.add("Fried Rice - 150");
item.add("Pizza - 250");
item.add("Burger - 120");
item.add("Pasta - 180");
item.add("Coffee - 80");

qty = new TextField();

add = new Button("Add Item");
next2 = new Button("Next");
previous1 = new Button("Previous");
clear = new Button("Clear");

orderList = new List();

p2.add(l3);
p2.add(category);

p2.add(l4);
p2.add(item);

p2.add(l5);
p2.add(qty);

p2.add(add);
p2.add(clear);

p2.add(previous1);
p2.add(next2);

p2.add(new Label("Selected Items"));
p2.add(orderList);

Panel p3 = new Panel(new BorderLayout());

bill = new TextArea();

generate = new Button("Generate Bill");
previous2 = new Button("Previous");

Panel bp = new Panel();
bp.add(previous2);
bp.add(generate);

p3.add(bill, BorderLayout.CENTER);
p3.add(bp, BorderLayout.SOUTH);

cards.add(p1, "Customer");
cards.add(p2, "Food");
cards.add(p3, "Bill");

f.add(cards);

next1.addActionListener(this);
next2.addActionListener(this);
previous1.addActionListener(this);
previous2.addActionListener(this);
add.addActionListener(this);
generate.addActionListener(this);
clear.addActionListener(this);

```



```

f.setSize(600,500);
f.setVisible(true);

f.addWindowListener(new WindowAdapter()
{
    public void windowClosing(WindowEvent e)
    {
        f.dispose();
    }
});
}

public void actionPerformed(ActionEvent e)
{
    if(e.getSource()==next1)
    {
        if(name.getText().equals("") || table.getText().equals(""))
        {
            bill.setText("Please enter customer name and table number.");
            cl.show(cards,"Customer");
        }
        else
        {
            cl.show(cards,"Food");
        }
    }

    if(e.getSource()==add)
    {
        if(qty.getText().equals(""))
        {
            return;
        }

        try
        {
            int q = Integer.parseInt(qty.getText());

            if(q <= 0)
            {
                return;
            }

            String selected = item.getSelectedItem();

            String[] parts = selected.split("-");

            String foodName = parts[0];
            double price = Double.parseDouble(parts[1]);

            double cost = price * q;

            subtotal = subtotal + cost;

            orderList.add(foodName + " x " + q + " = Rs." + cost);

            qty.setText("");
        }
        catch(Exception ex)
        {

```

```

        qty.setText("");
    }
}

if(e.getSource()==clear)
{
    orderList.removeAll();
    subtotal = 0;
}

if(e.getSource()==next2)
{
    if(orderList.getItemCount()==0)
    {
        return;
    }

    cl.show(cards,"Bill");
}

if(e.getSource()==previous1)
{
    cl.show(cards,"Customer");
}

if(e.getSource()==previous2)
{
    cl.show(cards,"Food");
}

if(e.getSource()==generate)
{
    double gst = subtotal * 0.05;
    double total = subtotal + gst;

    bill.setText("");

    bill.append("          FOOD ORDER BILL\n");
    bill.append("-----\n");
    bill.append("Customer Name : " + name.getText() + "\n");
    bill.append("Table Number  : " + table.getText() + "\n");
    bill.append("-----\n");

    String orders[] = orderList.getItems();

    for(int i=0;i<orders.length;i++)
    {
        bill.append(orders[i] + "\n");
    }

    bill.append("-----\n");
    bill.append("Subtotal      : Rs." + subtotal + "\n");
    bill.append("GST (5%)      : Rs." + gst + "\n");
    bill.append("Final Amount  : Rs." + total + "\n");
    bill.append("-----\n");
    bill.append("          Thank You!\n");
}
}

public static void main(String args[])
{

```

```

    new FoodOrdering();
}

```

OUTPUT:

Online Food Ordering and Billing System

Customer Name

Bala

Table Number

5

Next

Online Food Ordering and Billing System

Food Category

Beverages

Food Item

Coffee - 80

Quantity

Add Item

Clear

Previous

Next

Selected Items

Biryani x 2 = Rs.440.0
Coffee x 2 = Rs.160.0

Online Food Ordering and Billing System

FOOD ORDER BILL

Customer Name : Bala

Table Number : 5

Biryani x 2 = Rs.440.0

Coffee x 2 = Rs.160.0

Subtotal : Rs.600.0

GST (5%) : Rs.30.0

Final Amount : Rs.630.0

Thank You!